Summary in Graph

Exam Summary (GO Classes GATE CS | Relations | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	7 (0 + 7)
Total Marks:	14 (0 + 14)
Exam Duration:	21 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7

* Average Stats based on first-attempt stats of Top 10% (30) out of 304 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 2 Penalty: 0 Set Theory & Algebra

A binary relation R over a set A is called a "GO Relation" if for all $\mathrm{x}, \mathrm{y}, \mathrm{z} \in A$, if xRy and xRz, then yRz.

Which of the following is/are true about a relation R?

A. If R is a reflexive and GO relation then R is symmetric.
B. If R is a reflexive and GO relation then R is transitive.
C. If R is a GO relation then R is reflexive.

D. If R is an equivalence relation then R is a GO relation.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A binary relation R over a set A is called surjective if the following statement is true about R :

$\forall \mathrm{b} \in \mathrm{A}. \exists \mathrm{a} \in \mathrm{A}. \mathrm{aRb}$

Which of the following statements is(are) true?

A. A binary relation R over a set A is an equivalence relation if R is surjective, symmetric, and transitive.
B. If a binary relation R over a set A is an equivalence relation then R is surjective.
C. If a binary relation R over a set A is reflexive then R is surjective.
D. If a binary relation R over a set A is surjective then R is reflexive.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let R and T be binary relations over the same set A. We'll say that R is "no stronger than" T if the following statement is true:

$$\forall a \in \mathrm{A}. \forall \mathrm{b} \in \mathrm{A}. (\mathrm{aRb} \rightarrow \mathrm{aTb})$$

Let R and T be binary relations over the same set A where R is "no stronger than" T.

Then which of the following is/are true?

A. If R is an equivalence relation, then T is an equivalence relation.
B. If T is an equivalence relation, then R is an equivalence relation.
C. If R is a reflexive relation, then T is a reflexive relation.
D. If T is a reflexive relation, then R is a reflexive relation.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let A, B be two finite, non-empty sets, with cardinality of A, B as m, n respectively.

Let $\mathrm{f}: \mathrm{A} \rightarrow \mathrm{B}$ be a function with domain A and co-domain B. Let r be the cardinality of range of f.

We define a relation R on A, defined by $(x, y) \in \mathrm{R} \Leftrightarrow \mathrm{f}(x)=\mathrm{f}(y)$.

Which of the following statements is true?

A. R is not an equivalence relation.
B. R is an equivalence relation having r equivalence classes.
C. R is an equivalence relation having m equivalence classes.
D. R is an equivalence relation having n equivalence classes.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A relation R is said to be circular if aRb and bRc together imply cRa.

Which of the following options is/are correct?

A. If a relation S is circular, then S is a transitive relation.
B. If a relation S is circular and symmetric, then S is a transitive relation.
C. If a relation S is reflexive and circular, then S is a symmetric relation.
D. If a relation S is transitive and circular, then S is a symmetric relation.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A binary relation R over a set A is called total iff for any $x \in A$ and $y \in A$, at least one of xRy or yRx is true.

Which of the following is/are true?

A. A binary relation R over a set A is a total order iff it is a partial order and it is total.
B. Every total relation is reflexive.
C. If a relation S is total $\&$ symmetric, then S is an equivalence relation.
D. Every total relation is transitive.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 04sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A Relation R on set S is said to be Serial if $\forall x \exists y\ R(x, y)$.

Let T be a symmetric and transitive relation on a non-empty set A, then which of the following is/are true?

A. T is reflexive.
B. For any element $x \in A$, if xTy for some $y \in A$, then xRx is necessarily true.
C. If T is serial also, then T is reflexive.
D. Every equivalence relation is serial.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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